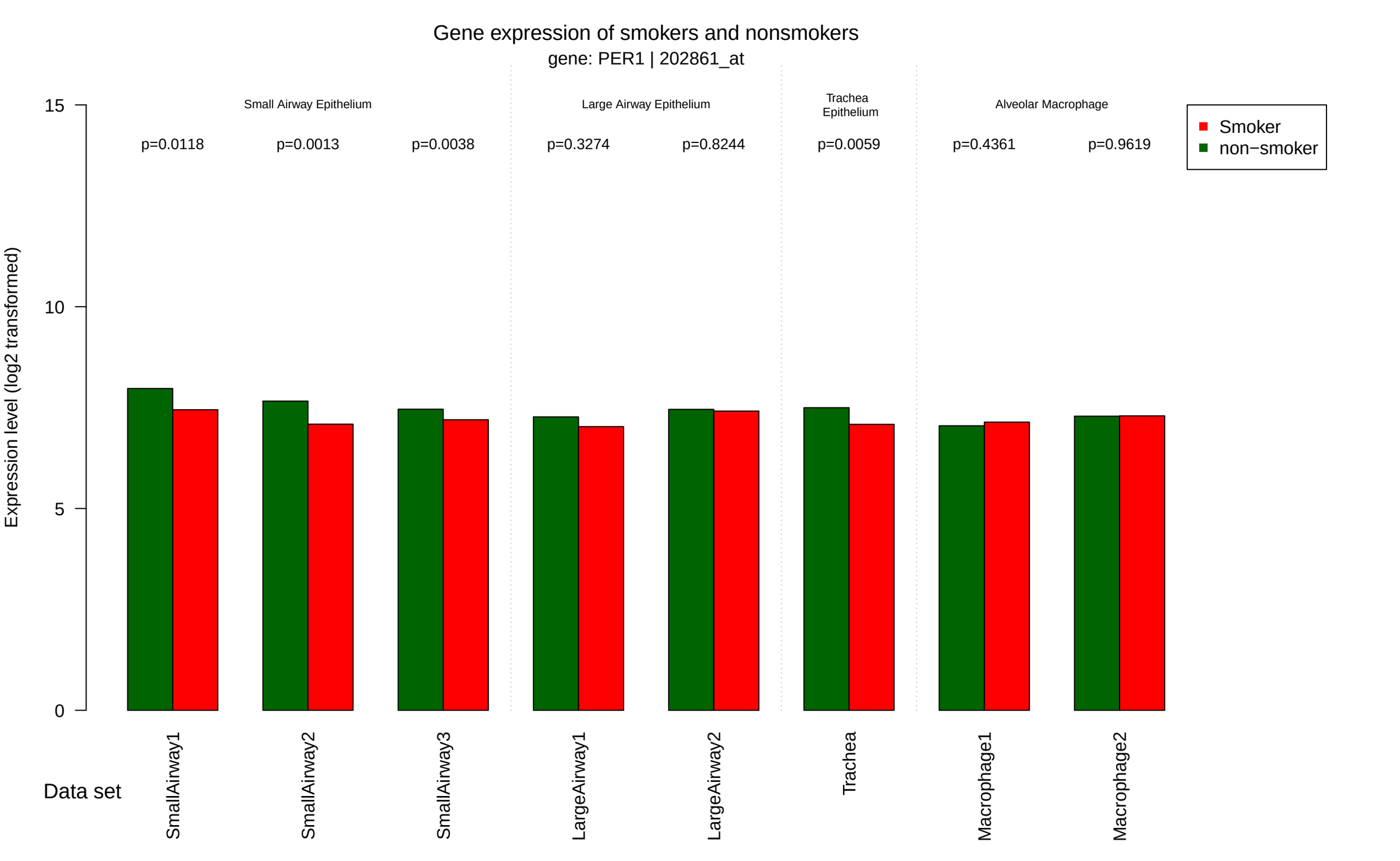
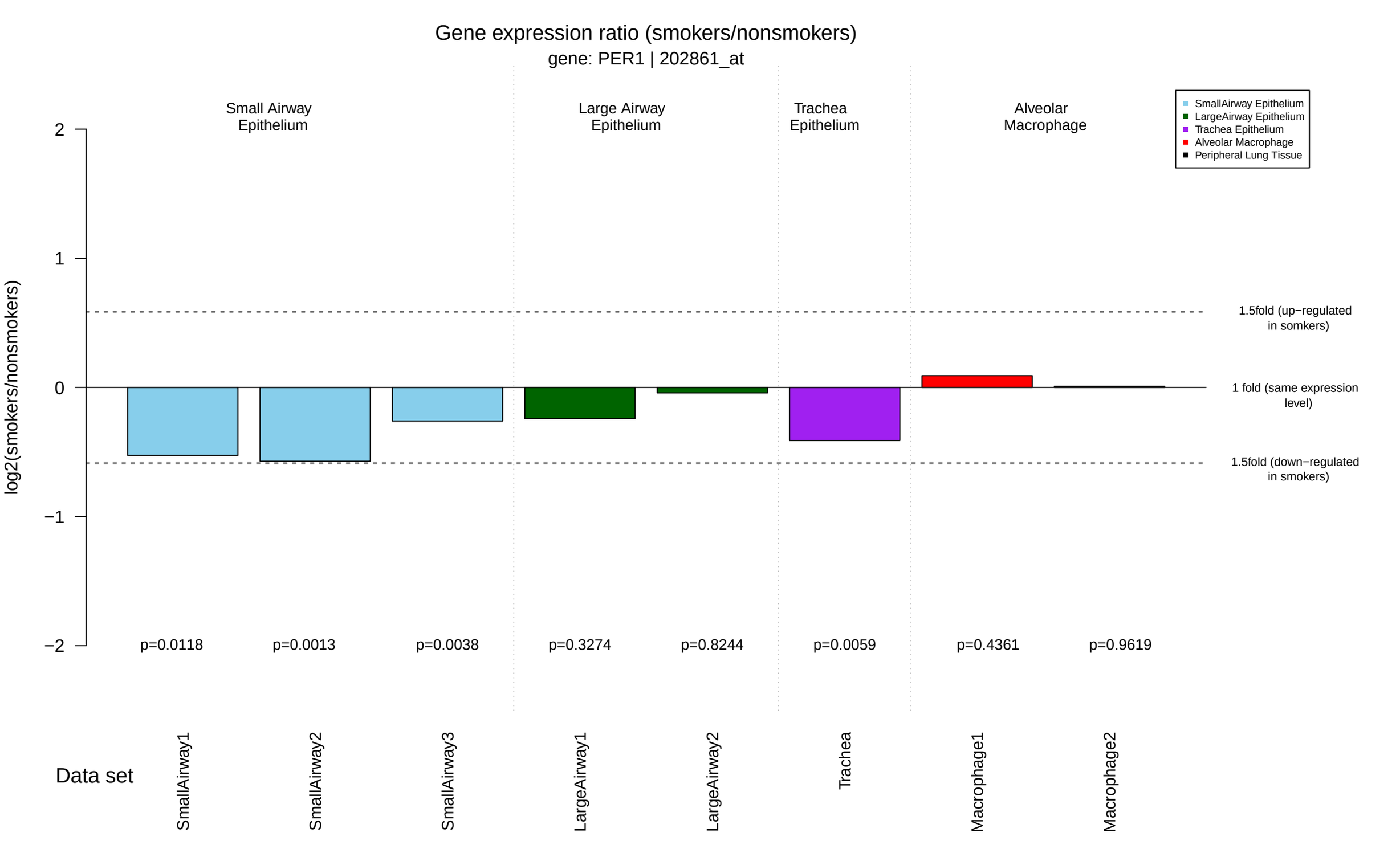
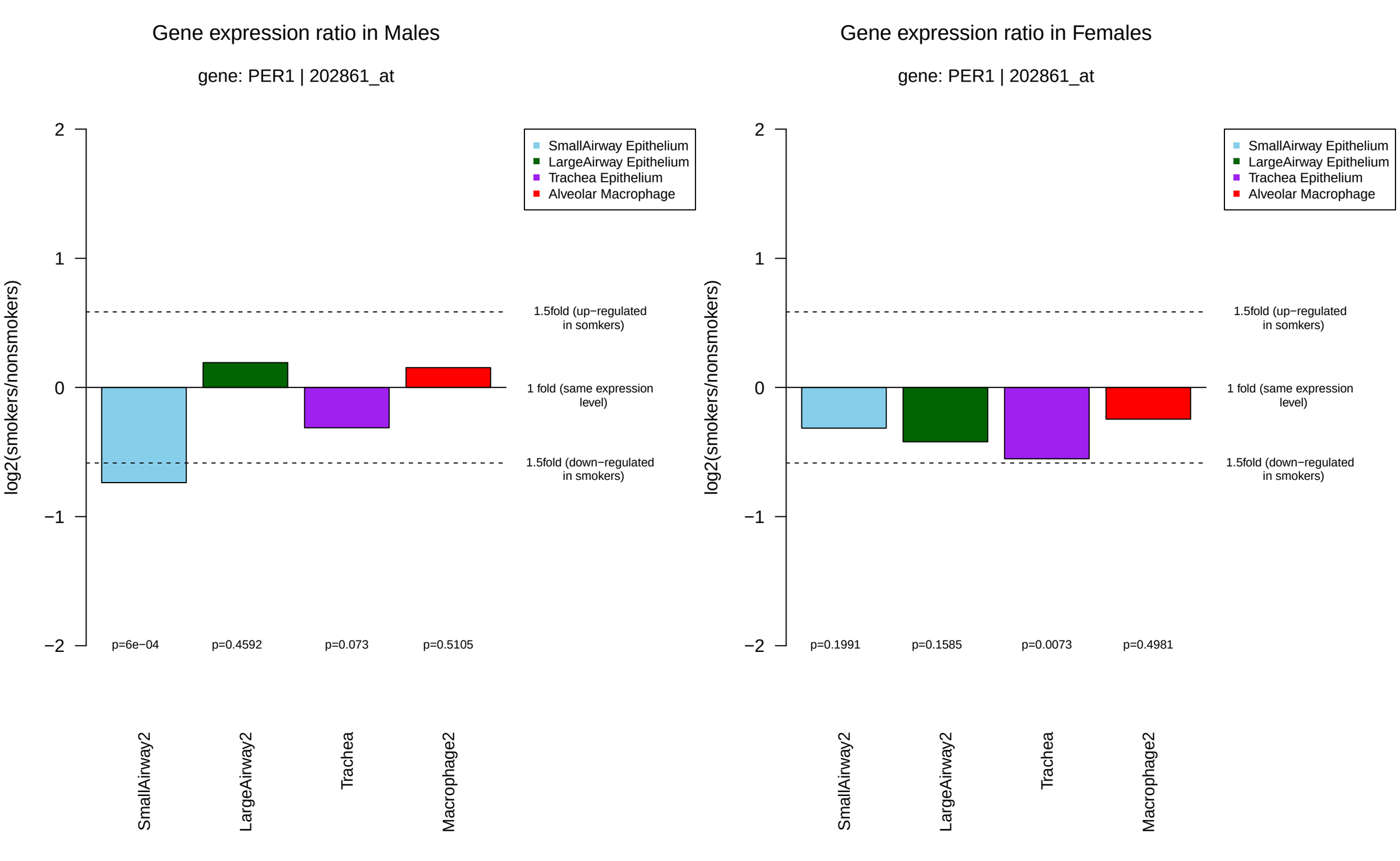
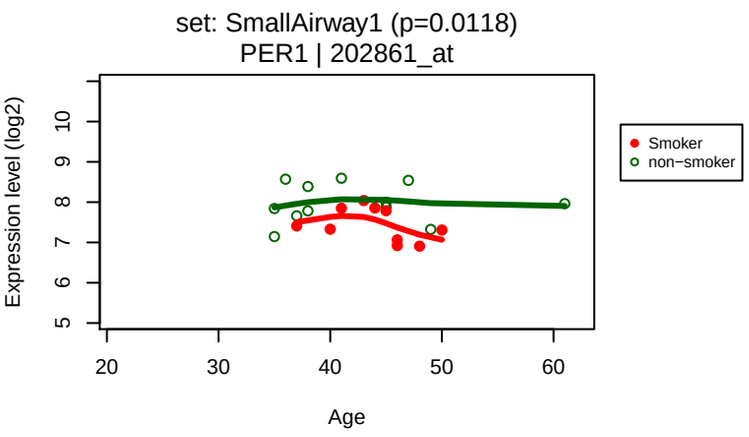
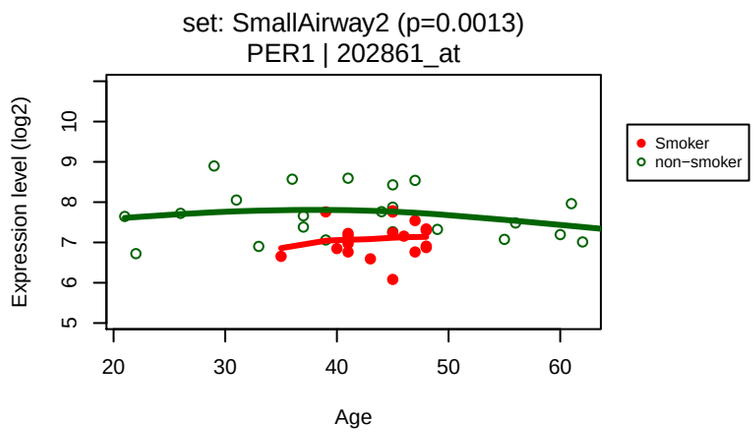
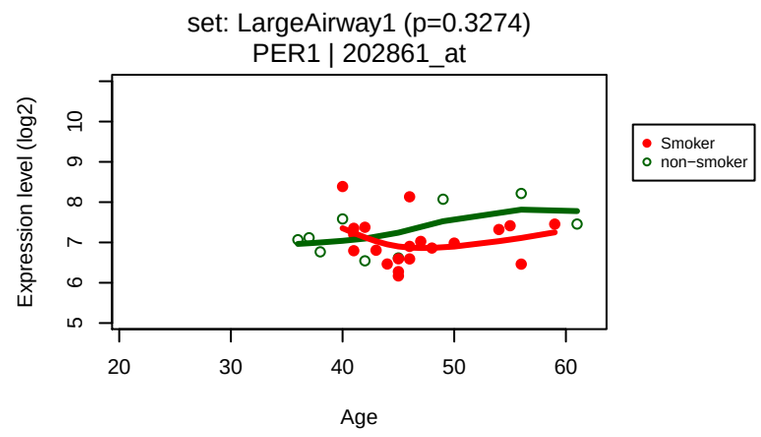

Gene expression of smokers and nonsmokers
gene: PER1 | 202861_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0118
p=0.0013
p=0.0038
p=0.3274
p=0.8244
p=0.0059
p=0.4361
p=0.9619
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

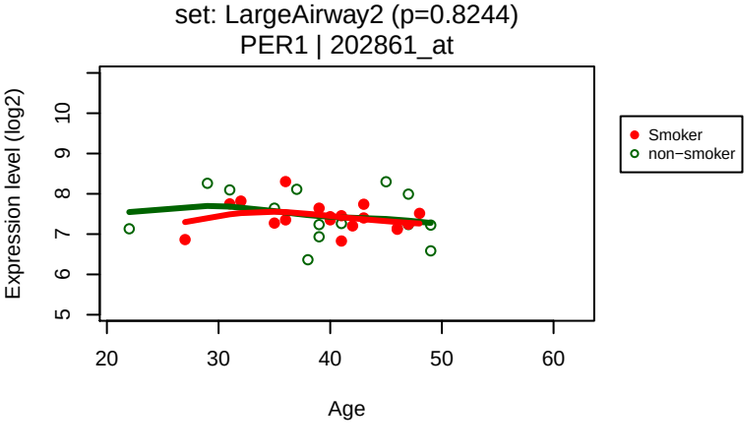

Gene expression ratio (smokers/nonsmokers)
gene: PER1 | 202861_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0118
p=0.0013
p=0.0038
p=0.3274
p=0.8244
p=0.0059
p=0.4361
p=0.9619
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

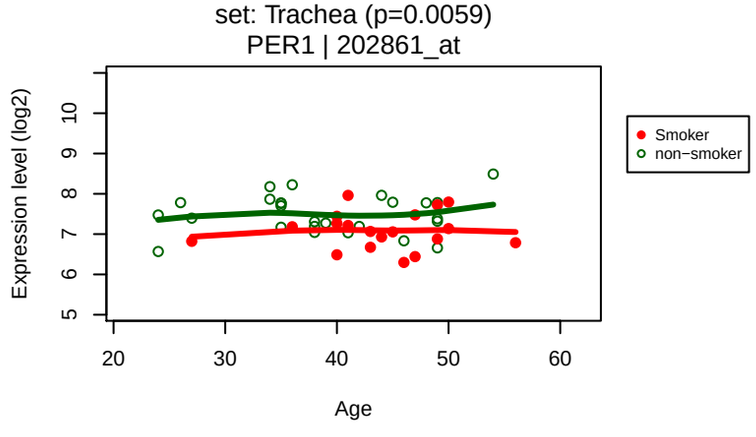

Gene expression ratio in Males
gene: PER1 | 202861_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=6e−04
p=0.4592
p=0.073
p=0.5105
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PER1 | 202861_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1991
p=0.1585
p=0.0073
p=0.4981
SmallAirway2
LargeAirway2
Trachea
Macrophage2

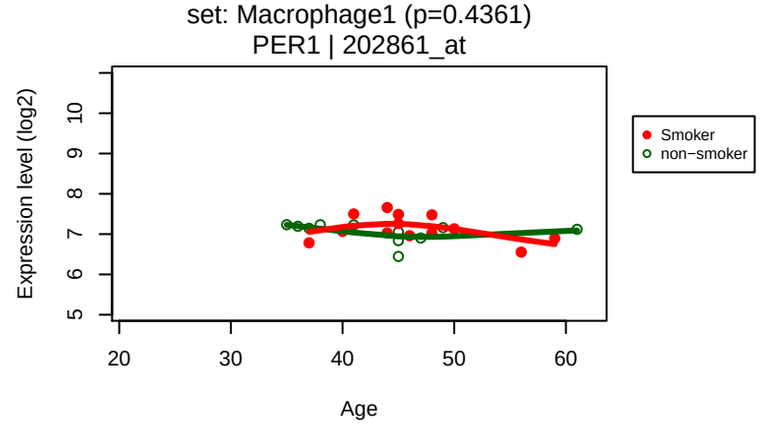

set: SmallAirway1 (p=0.0118)
PER1 | 202861_at
Expression level (log2)
Age
Smoker
non−smoker

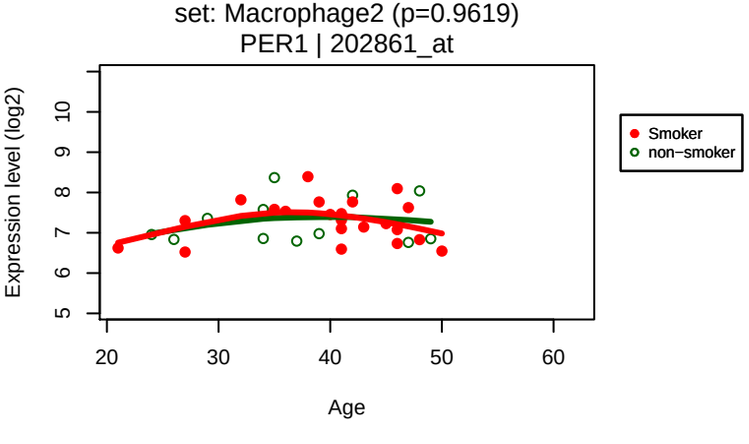

set: SmallAirway2 (p=0.0013)
PER1 | 202861_at
Expression level (log2)
Age
Smoker
non−smoker

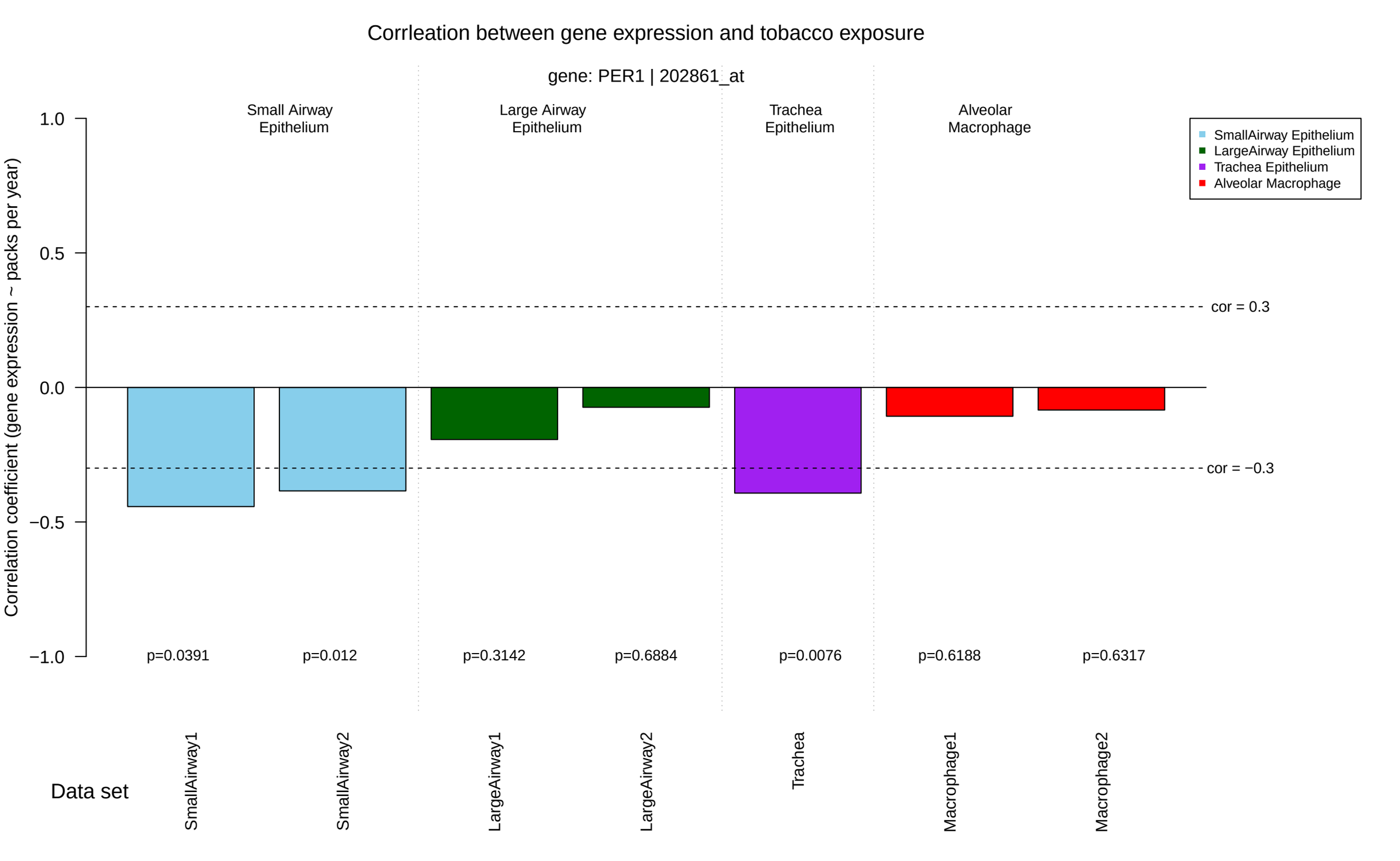

set: LargeAirway1 (p=0.3274)
PER1 | 202861_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8244)
PER1 | 202861_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0059)
PER1 | 202861_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4361)
PER1 | 202861_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9619)
PER1 | 202861_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PER1 | 202861_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0391
p=0.012
p=0.3142
p=0.6884
p=0.0076
p=0.6188
p=0.6317
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2